Word Games Hub & Leaderboard

End-to-End Full-Stack Architecture & Implementation Plan

1. Executive Summary & Architectural Vision

This document outlines the end-to-end architecture, normalization engine, database schema, accessibility standards, and implementation roadmap for a real-time multiplayer daily word game tracking platform. Friends can asynchronously log their daily results across disparate word and trivia games (e.g., **Wordle**, **Word500**, **Krillion**, Connections, Contexto) and compete on dynamically recalculated **Daily** and **All-Time** leaderboards where every game carries equal weight.

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2. Core Engineering Challenge: The Equal-Weight Normalization Engine

Different word games measure success using incompatible scales and directions:

- **Wordle:** Attempt-based (lower is better, 1 to 6 tries, or X/fail).
- **Word500:** Attempt-based (lower is better, 1 to 8 tries to deduce the 5-letter word via count-only feedback, or X/fail).
- **Krillion:** Points-based (higher is better, 0 to 700 points across 7 timed prompts where each point equals 10 meters of depth).

To ensure **every game is weighted equally** and scores update dynamically as users enter games asynchronously throughout the day:

1. **Standardized 0–100 Normalized Scale (`normalized_score`):** Every raw score is converted on write (via a deterministic backend function/trigger) to a value between `0.00` and `100.00`.
2. **Scoring Formulas by Game Type:**
    - **Wordle (Max 6 tries, lower is better):**
        - 1 try = `100.00` pts
        - 2 tries = `90.00` pts
        - 3 tries = `80.00` pts

        - 4 tries = `65.00` pts
        - 5 tries = `50.00` pts
        - 6 tries = `35.00` pts
        - Failed (X) = `10.00` pts (participation credit to reward playing vs. skipping)
    - **Word500 (Max 8 tries, lower is better):**
        - Formula for solved: `100 - ((attempts - 1) * 11)` (1 try = `100`, 2 = `89`, 3 = `78`, 4 = `67`, 5 = `56`, 6 = `45`, 7 = `34`, 8 = `23`)
        - Failed (X) = `10.00` pts
    - **Krillion (0 to 700 max points, higher is better):**
        - Formula: `ROUND((raw_score / 700.0) * 100, 2)`
        - Example: A raw score of `325` yields `(325 / 700) * 100 = 46.43` normalized points.
3. **Asynchronous Partial Submissions:**
    - A player's **Daily Composite Score** is the **sum** of normalized scores for all games submitted that day (max `N * 100` for `N` active games), alongside a secondary **Average Per Game Played** metric so players who have only finished 1 of 3 games by midday can still compare their efficiency fairly.
    - As soon as a player enters a single game score, the database recalculates their daily total and broadcasts the updated leaderboard to all connected clients.

Score Normalization Reference Table

Game	Metric / Scale	Raw Score	Formula / Mapping	Normalized Score
Wordle	Attempts (Lower Better)	1–6 Tries	Direct Step Scale (100, 9...	35.00 – 100.00
Wordle	Failed	X	Flat Participation Credit	10.00
Word500	Attempts (Lower Better)	1–8 Tries	`100 - ((attempts - 1…`	23.00 – 100.00
Word500	Failed	X	Flat Participation Credit	10.00
Krillion	Points (Higher Better)	0–700 Pts	`ROUND((raw_score / 7…`	0.00 – 100.00

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3. Recommended 100% Free Tech Stack & Database Selection

Why PostgreSQL (Relational) Beats NoSQL / MySQL Here

- **PostgreSQL (via Supabase Free Tier)** is the optimal choice over NoSQL (MongoDB/Firebase) and standard MySQL:
    - **Window Functions & Ranking:** Native support for `DENSE_RANK() OVER (ORDER BY SUM(normalized_score) DESC)` makes tie-breaking, daily aggregations, and all-time rollups effortless and fast.
    - **ACID & Uniqueness Constraints:** Enforces strict `UNIQUE (user_id, game_id, play_date)` constraints at the storage layer so double-tapping "Submit" never duplicates scores.
    - **Built-in Real-time Subscriptions:** Supabase provides PostgreSQL Change Data Capture (CDC) over WebSockets out of the box, pushing live leaderboard shifts to all 15+ concurrent users with zero polling overhead.

Full Free-Tier Stack Architecture

Layer	Recommended Technology	Free Tier Allocation & Utility
Frontend & API	Next.js 15 (App Router, TypeScript)	Hosted on **Vercel** / **Cloudflare Pages** (Unlimited bandwidth/builds)
Styling & UI	Tailwind CSS + Radix UI / shadcn/ui	Built-in WAI-ARIA primitives & full keyboard navigation
Database & Auth	Supabase (Managed PostgreSQL)	500 MB DB, Auth (Magic Link / OAuth), RLS Security Engine
Realtime Sync	Supabase Realtime Engine	WebSockets (200 concurrent connections, 50k MAU)

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4. Use Cases & User Flows

- **UC-01: Frictionless Magic Link / OAuth Sign-In:** User joins the private friend group via invite code or Google login and sets a display name and high-contrast avatar.
- **UC-02: Quick-Tap Asynchronous Score Entry:** User opens the dashboard, sees cards for today's games (Krillion, Wordle, Word500), taps **Krillion**, types `325`, and

hits Save. The card flips to "Completed (46.43 pts)" and the Daily Leaderboard immediately animates their rank shift. Hours later, they tap **Wordle**, select `3 tries`, and their daily score increases by `80.00` pts to `126.43` pts.

- **UC-03: Share-Text Auto-Parser (Quality-of-Life Best Practice):** Instead of typing numbers manually, users can optionally paste their native emoji share block from Wordle/Word500/Krillion; a regex parser automatically extracts the attempt count/score and verifies no spoilers are shown.
- **UC-04: Viewing Daily vs. All-Time Leaderboards:** User toggles between **Today's Standings** (filtering by SAST / UTC+2 local date) and **All-Time Hall of Fame** (Cumulative Points, Win Rate %, Current Streak, and Per-Game Specialty Badges).
- **UC-05: Edit / Audit Window:** If a user mistypes `352` instead of `325`, they can edit today's score before midnight local time, logging an audit trail entry to keep friends honest.
- **UC-06: Admin Game Configuration:** Group admin can add a new daily game (e.g., Connections) by defining its `scoring_direction` (`LOWER_BETTER` or `HIGHER_BETTER`), `min_val`, and `max_val` without changing code.

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5. Functional Requirements (FR)

- **FR-1 (Authentication & Group Isolation):** Users must authenticate via Supabase Auth; leaderboard views are scoped to the friend group (`group_id`).
- **FR-2 (Game Registry):** System must store metadata and normalization rules for each supported game (`wordle`, `word500`, `krillion`, etc.).
- **FR-3 (Partial & Incremental Score Logging):** Users can submit or update individual game scores independently at any time during the 24-hour daily cycle.
- **FR-4 (Dynamic Normalization & Aggregation):** Every score insert/update must automatically compute `normalized_score` (0–100) and refresh both the Daily and All-Time leaderboard views.
- **FR-5 (Daily Timezone Cutoff):** Daily leaderboards must reset cleanly at `00:00:00` local timezone (configurable per group, defaulting to `Africa/Johannesburg` GMT+2).
- **FR-6 (All-Time Metrics):** The All-Time leaderboard must track **Total Normalized Points**, **Mean Score per Game Played** (for fairness if someone misses a day), **Daily Crowns Won**, and **Active Streak**.

- **FR-7 (Anti-Cheat & Validation):** Backend must reject out-of-range inputs (e.g., Wordle `< 1` or `> 6`, Krillion `< 0` or `> 700`, Word500 `< 1` or `> 8`).

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6. Non-Functional Requirements (NFR) & WCAG 2.1 AA Compliance

- **NFR-1 (Concurrency & Performance):** Support a minimum of **15–100 concurrent active users** submitting scores and viewing live updates simultaneously, with API p95 latency under `200ms` and real-time broadcast latency under `500ms`.
- **NFR-2 (WCAG 2.1 AA Color & Contrast Compliance):**
  - **Contrast Ratios:** All body text must maintain at least a **4.5:1** contrast ratio against its background; large headings and score badges must maintain at least **3:1** (targeting **7:1 AAA** for primary leaderboard tables).
  - **Colorblind-Safe Palette (Okabe-Ito Inspired):** Never rely on color alone to indicate rank changes or game status. Pair colors with explicit icons (`▲ +2`, `▼ -1`, `✓ Done`). Use high-contrast dark/light palettes:
    - Light Mode Background: `#F8FAFC` (Slate 50) with Primary Text `#0F172A` (Slate 900 — Contrast **17.4:1**).
    - Primary Accent: `#1D4ED8` (Blue 700) with White Text `#FFFFFF` (Contrast **8.6:1**).
    - Success / Completed Badge: `#047857` (Emerald 700) with `#FFFFFF` (Contrast **5.4:1**).
    - Warning / Partial Badge: `#B45309` (Amber 700) with `#FFFFFF` (Contrast **5.0:1**).
- **NFR-3 (WCAG Typography & Font Sizing):**
  - **Font Family:** Highly legible humanist/geometric sans-serif (**Inter** or **Atkinson Hyperlegible**) paired with **JetBrains Mono** (`font-variant-numeric: tabular-nums`) for aligned numeric columns.
  - **Font Sizes:** Base body font size strictly `16px` (`1rem`) minimum; mobile numeric inputs at least `18px` (`1.125rem`) to prevent iOS auto-zoom; touch targets minimum `48x48px`; line height `1.5` and support for `200%` browser text zoom without horizontal scrolling.
- **NFR-4 (Zero-Cost Operations):** Infrastructure must run within 100% free tiers without requiring credit card overages.
- **NFR-5 (Security):** PostgreSQL Row-Level Security (RLS) policies ensuring users can only `INSERT` or `UPDATE` rows where `auth.uid() = user_id`.

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7. PostgreSQL Database Schema & Normalization SQL

```
None
-- 1. Games Configuration Table
CREATE TABLE games (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  slug TEXT UNIQUE NOT NULL, -- 'wordle', 'word500', 'krillion'
  name TEXT NOT NULL,
  scoring_type TEXT NOT NULL CHECK (scoring_type IN ('ATTEMPTS_LOWER_BETTER',
'POINTS_HIGHER_BETTER')),
  min_score INT NOT NULL,
  max_score INT NOT NULL,
  is_active BOOLEAN DEFAULT true
);

-- 2. Daily Submissions Table
CREATE TABLE submissions (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id UUID NOT NULL REFERENCES auth.users(id) ON DELETE CASCADE,
  game_id UUID NOT NULL REFERENCES games(id) ON DELETE CASCADE,
  play_date DATE NOT NULL DEFAULT (CURRENT_TIMESTAMP AT TIME ZONE
'Africa/Johannesburg')::DATE,
  raw_score INT, -- NULL if failed (X) on attempt-based games
  is_failed BOOLEAN DEFAULT false,
  normalized_score NUMERIC(5,2) NOT NULL,
  updated_at TIMESTAMPTZ DEFAULT now(),
  UNIQUE(user_id, game_id, play_date)
);

-- 3. Daily Leaderboard View (Equal Weighting)
CREATE VIEW daily_leaderboard AS
SELECT
  s.play_date,
  u.id AS user_id,
  u.raw_user_meta_data->>'display_name' AS display_name,
  COUNT(s.game_id) AS games_played,
  SUM(s.normalized_score) AS total_daily_score,
```

```
  ROUND(AVG(s.normalized_score), 2) AS avg_game_score,
  DENSE_RANK() OVER (
    PARTITION BY s.play_date
    ORDER BY SUM(s.normalized_score) DESC, COUNT(s.game_id) DESC
  ) AS daily_rank
FROM submissions s
JOIN auth.users u ON u.id = s.user_id
GROUP BY s.play_date, u.id;

-- 4. All-Time Leaderboard View
CREATE VIEW all_time_leaderboard AS
SELECT
  u.id AS user_id,
  u.raw_user_meta_data->>'display_name' AS display_name,
  COUNT(s.id) AS total_games_logged,
  COUNT(DISTINCT s.play_date) AS days_active,
  SUM(s.normalized_score) AS cumulative_normalized_score,
  ROUND(AVG(s.normalized_score), 2) AS career_average_score,
  DENSE_RANK() OVER (
    ORDER BY SUM(s.normalized_score) DESC
  ) AS all_time_rank
FROM submissions s
JOIN auth.users u ON u.id = s.user_id
GROUP BY u.id;
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8. Step-by-Step Implementation Roadmap (4 Phases)

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None
+-------------------------------------------------------------------------------
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|                            IMPLEMENTATION TIMELINE
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+------------------------+-----------------------+-----------------------------
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| Phase 1: DB & Logic    | Phase 2: Next.js Auth | Phase 3: UI & Realtime
| Phase 4: Audit & Launch  |
```

```
| - Supabase DDL         | - App Router Setup    | - Bento Grid Input Cards
| - Share Parser           |
| - Seed Core Games      | - Auth & Navigation   | - Supabase CDC WebSockets
| - WCAG AA Audit          |
| - Normalization Trigger| - WCAG CSS Variables  | - Animated Leaderboard
| - Vercel Deployment      |
+------------------------+-----------------------+----------------------------
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```

Phase 1: Database & Normalization Setup (Day 1)

1. Provision a free **Supabase** project and execute the DDL script for `games`, `submissions`, and RLS policies.
2. Seed the `games` table with **Wordle** (`1..6`), **Word500** (`1..8`), and **Krillion** (`0..700`).
3. Create a PostgreSQL `BEFORE INSERT OR UPDATE` trigger function that automatically calculates `normalized_score` from `raw_score` so client code can never tamper with point weighting.

Phase 2: Next.js Scaffolding & Auth (Day 2)

1. Initialize Next.js 15 with TypeScript, Tailwind CSS, and `@supabase/ssr`.
2. Configure **Atkinson Hyperlegible** / **Inter** and **JetBrains Mono** via `next/font/google` and set WCAG AA CSS variables in `globals.css`.
3. Build the Group Login page and a persistent bottom/top navigation bar switching between **Log Scores**, **Daily Leaderboard**, and **All-Time Standings**.

Phase 3: Quick-Entry Game Cards & Realtime Leaderboards (Days 3–4)

1. Build the **Quick-Tap Game Bento Grid**:
    - **Wordle & Word500 Cards:** Render 1-tap pill buttons (`1`, `2`, `3`, `4`, `5`, `6`, `X` for Wordle; `1..8`, `X` for Word500) so users don't even have to open a keyboard on mobile.
    - **Krillion Card:** Render a large `48px`-tall numeric input (`inputmode="numeric"`, `0–700`) with instant normalized point preview (e.g., typing `325` previews `= 46.43 pts`).
2. Wire **Supabase Realtime** on the `submissions` table so whenever a friend logs a game, the leaderboard re-sorts smoothly using Framer Motion layout animations (`prefers-reduced-motion` respected).

Phase 4: Accessibility Audit, Share-Parser & Deployment (Day 5)

1. Add the **Clipboard Share Parser** to auto-read pasted emoji grids from Wordle/Word500.
2. Run **Lighthouse Accessibility** and **axe DevTools** audits to verify 100% WCAG 2.1 AA compliance (focus rings, `aria-live="polite"` on leaderboard rank updates, contrast ratios).
3. Deploy to **Vercel** (Free Tier) and share the invite URL with your group.